Installation Guide

From zero to your first notebook, step by step

> ! macOS or Windows?
>
> This guide is different for macOS and Windows. Choose your system in the tabs below, and the whole page follows.

Before you start

You don't need any prior knowledge. Follow the steps in order and don't skip any.

You need:

- Your **laptop** (Windows or macOS) with internet access
- The **password** of your laptop, because installing software sometimes asks for it

At the end, you will have everything you need for the tutorials.

The big picture: five things you will use

What	In simple words	Why we need it
Python	A programming language	We write our programs in Python
VS Code	An app to write code, like Word for text	This is where you write and run your code
Terminal	A window where you type commands instead of clicking	We use it to install things
uv	A small helper tool	It installs Python and everything Python needs
Notebook	A document that mixes text and code you can run	All tutorials are notebooks

How they fit together: you open **VS Code**, use its **terminal** to let **uv** install **Python**, and then work in **notebooks** inside VS Code.

Step 1: Install VS Code

Where: your web browser

Why: VS Code is the app you will work in for the whole course. Apps like this are called IDEs (Integrated Development Environments): one place to write code, run it and find mistakes.

Do: Go to code.visualstudio.com and download VS Code for your system.

macOS

1. Open the downloaded file. A file called **Visual Studio Code** appears.
2. Drag it into your **Applications** folder.
3. Open it from Applications. If your Mac asks whether you want to open an app from the internet, click **Open**.

Windows

1. Open the downloaded installer.
2. Accept the agreement and keep the default settings. Make sure **"Add to PATH"** is ticked.
3. Click **Install**, then **Finish**.

Check: VS Code opens and shows a welcome page.

Step 2: Add the Python and Jupyter extensions

Where: VS Code

Why: Out of the box, VS Code doesn't know Python or notebooks. Extensions are small add-ons that teach it.

Do:

1. Click the **Extensions** icon on the left side (four small squares).
2. Search for **Python** and click **Install** on the one by Microsoft.
3. Search for **Jupyter** and click **Install** on the one by Microsoft.

Check: Both extensions show "Installed" or a gear icon instead of an Install button.

Step 3: Create your course folder

Where: Finder (macOS) or File Explorer (Windows)

Why: Everything for this course lives in one folder. That keeps things tidy, and Python works best when each project has its own folder.

macOS

Open **Finder**, go to your **home folder** (the one with the house icon) and create a new folder called `programming-course`.

Windows

Open **File Explorer**, click into the address bar at the top, type `%USERPROFILE%` and press **Enter**. This opens your user folder. Create a new folder there called `programming-course`.

Please **don't** put it in a OneDrive folder. Syncing can break the setup.

Do in VS Code: Click **File → Open Folder...**, select `programming-course` and click **Open**. If VS Code asks whether you trust the authors of the files, click **Yes, I trust the authors**.

Check: On the left side of VS Code, you see the name `PROGRAMMING-COURSE` at the top.

Step 4: Open the terminal

Where: VS Code

Why: The terminal is where you type commands. You type a command, press **Enter**, and the computer answers with text. We only need a few commands, and you can copy them from this page.

Do: In the menu bar, click **Terminal → New Terminal**. A panel opens at the bottom of VS Code.

macOS

The terminal shows a line ending with `%`. That's where you type.

Windows

The terminal shows a line starting with `PS` and ending with `>`. `PS` means **PowerShell**, which is exactly what we need. If you don't see `PS`, click the small arrow next to the **+** in the terminal panel and choose **PowerShell**.

Good to know:

- Copy a command from this page, paste it into the terminal and press **Enter**.
- Paste with **Cmd + V** (macOS) or **Ctrl + V** or a right-click (Windows).
- Some commands take a few seconds. Wait until the line with `%` or `>` is back before typing the next one.

Check: You see a terminal at the bottom of VS Code, and its line mentions `programming-course`.

Step 5: Install uv

Where: the terminal in VS Code

Why: uv is a helper that installs Python for you and keeps each project clean. You only install it once.

Do: Copy the command for your system and press **Enter**.

macOS

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Windows

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

⚠ Restart required

The terminal only learns about new tools when VS Code starts fresh.

macOS

Quit VS Code with **Cmd + Q**. Closing the window is not enough.

Windows

Close **all** VS Code windows.

Open VS Code again. **Check:** The top left must show `PROGRAMMING-COURSE` again. If it doesn't, click **File → Open Recent → programming-course**. Then open a new terminal with **Terminal → New Terminal**.

Check: Type this and press **Enter**:

```
uv --version
```

You should see something like `uv 0.12.0`. The exact number doesn't matter.

Step 6: Install Python

Where: the terminal in VS Code

Why: This installs the Python language itself. Everyone in the course uses the same version, so the code works the same for all.

Do:

```
uv python install 3.13
```

Check: The terminal says that Python 3.13 was installed, or that it is already installed. Both are fine.

Step 7: Turn your folder into a Python project

Where: the terminal in VS Code, inside `programming-course`

Why: This tells uv: "this folder is a Python project". uv creates a few files it needs to keep track of things.

Check first: The terminal must be inside your course folder. Type this and press **Enter**:

macOS

```
pwd
```

Windows

```
Get-Location
```

The answer must end with `programming-course`. If it doesn't: close the terminal (the bin icon in the terminal panel), open the folder with **File → Open Recent → programming-course**, and open a new terminal with **Terminal → New Terminal**.

Do: Run this command:

```
uv init --no-package --python 3.13
```

This creates a simple project with a `main.py` and Python 3.13 as its minimum version. (`--no-package` matters: since uv 0.12, plain `uv init` creates a `src/` folder without `main.py` instead.)

New files appear on the left in VS Code. You don't need to touch them:

- `main.py`: a tiny example program

- `pyproject.toml`: the list of what your project needs
- `.python-version`: which Python version the project uses

Check: Run the example program:

```
uv run python main.py
```

You should see `Hello from programming-course!`. Congratulations, you just ran your first Python program!

Step 8: Add notebook support

Where: the terminal in VS Code

Why: Notebooks need two extra helpers. The first one lets notebooks use the Python of your project, the second one adds the notebook tools.

Do:

```
uv add --dev ipykernel
```

```
uv add jupyter
```

Check: The terminal lists a lot of installed packages and ends without an error. You see a folder `.venv` on the left. That's your project's own Python environment.

Step 9: Your first notebook

Where: VS Code

Why: This checks that everything works together, before you start with the tutorials.

Do:

1. Press **Cmd + Shift + P** (macOS) or **Ctrl + Shift + P** (Windows). A search bar opens at the top.
2. Type `Create: New Jupyter Notebook` and press **Enter**.
3. Click **Select Kernel** in the top right corner. The kernel is the Python that runs your notebook.
4. Choose **Python Environments**, then the one that says **.venv** and **Python 3.13**. If you don't see Python Environments, click **Select Another Kernel...** first.
5. Click into the first cell, type `print("Hello")` and press **Shift + Enter**.

Check: Below the cell, you see `Hello`. Everything works!

If VS Code asks to install extensions at any point, click **Install**.

Step 10: Open a tutorial

Where: your web browser, then VS Code

Do:

1. Open a tutorial on the course website.
2. On the right side under **Other Formats**, click **Jupyter** to download the notebook.
3. Move the downloaded file into your `programming-course` folder.
4. Open it in VS Code by clicking it on the left.
5. If VS Code asks for a kernel, choose the same **.venv** with **Python 3.13** as in Step 9.

You're ready for the tutorials!

Checklist

- ☐ VS Code is installed and opens
- ☐ The Python and Jupyter extensions are installed
- ☐ You have a folder `programming-course` and opened it in VS Code
- ☐ `uv --version` shows a version number
- ☐ `uv run python main.py` shows `Hello from programming-course!`
- ☐ A notebook cell with `print("Hello")` shows `Hello`

Every next session

You don't need to install anything again. Just:

1. Open VS Code.
2. Click **File → Open Recent → programming-course**.
3. Open the notebook you want to work on.

Something doesn't work?

Problem	What to do
`uv` is "not found" or "not recognized"	Restart VS Code as in Step 5 (macOS: **Cmd + Q**, Windows: close all windows). On Windows, check that the terminal says `PS`.
`main.py` can't be found	Your terminal is probably in the wrong folder. Check the folder and the terminal as in Step 7, then run `uv`

Problem	What to do
	`init --no-package --python 3.13` again in the right folder. If uv then says the project is already initialized, see the next row. If `uv init` ran in a different folder by mistake: don't delete anything, ask us.
uv says the project is already initialized	Your project already exists, don't run `uv init` again. If you see `main.py`, continue with `uv run python main.py`. If there is a `src/` folder, see the row below. If neither exists, create a new file `main.py` on the left (New File icon) with the line `print("Hello from programming-course!")`, save it and run `uv run python main.py`.
There is a folder `src/` but no `main.py`	You ran `uv init` without `--no-package`. That's fine: test with `uv run programming-course` instead. It should print `Hello from programming-course!`. Then continue with Step 8.
uv says the Python version is incompatible with the project	Open `pyproject.toml` in VS Code and change the line `requires-python` to `requires-python = ">=3.13"`. Save the file, then run `uv python pin 3.13` and `uv sync`.
Windows says scripts are disabled	Run `Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process`, then repeat Step 5.
`.venv` doesn't appear under Select Kernel	Press **Cmd/Ctrl + Shift + P**, type `Developer: Reload Window`, press **Enter**, try again.
The notebook shows a different Python than 3.13	Click on the kernel name in the top right and choose the **.venv** one.
`uv self update` fails on a Mac	If you installed uv with Homebrew, use `brew upgrade uv` instead.
Anything else	Ask us! Guessing usually takes longer than asking.

Later in the course: adding packages

Sometimes a tutorial needs an extra package, for example `pandas`. Then run this in the terminal of your course folder:

```
uv add pandas
```